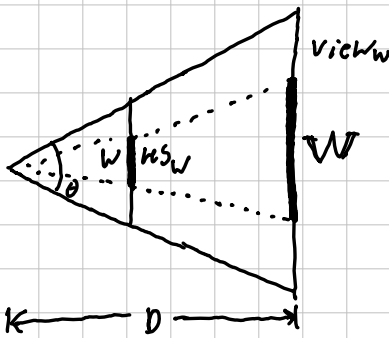


## how to determine fov



$$\tan\left(\frac{\theta_h}{2}\right) = \frac{\text{view}_w/2}{D}$$

$$\frac{w}{\text{res}_w} = \frac{W}{\text{view}_w}$$

$$\theta_h = 2 \tan^{-1}\left(\frac{\text{view}_w}{2D}\right)$$

$$= 2 \tan^{-1}\left(\frac{\text{res}_w W}{w}\right)$$

OR

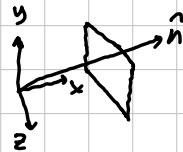
$$f = \frac{w \cdot D}{W}$$

$$\theta_h = 2 \tan^{-1}\left(\frac{\text{res}_w}{2f}\right)$$

## ray projection

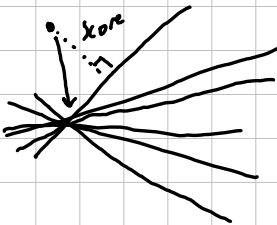
$$c_x = \frac{\text{res}_w}{2}, \quad c_y = \frac{\text{res}_h}{2}$$

$$\vec{n} = \left\langle \frac{x - c_x}{f}, \frac{y - c_y}{f}, 1 \right\rangle, \quad \hat{n} = \frac{\vec{n}}{\|\vec{n}\|}$$



$$\hat{n}_{\text{world}} = n_x \hat{i} + n_y \hat{j} + n_z \hat{k} \quad \leftarrow \text{camera basis vectors}$$

## ray intersection



## dist from point to ray



$$x = \|p - a\| \sin \theta$$

$$\|(p - a) \times (b - a)\| = \|p - a\| \|b - a\| \sin \theta$$

$$x = \frac{\|(p - a) \times (b - a)\|}{\|b - a\|}$$

## gradient descent

find  $x, y, z$  to minimize  $\sum \text{dist}$